

Practical Questions

1. Execute DDL statements

- i) Create a table Student with fields (RollNo,Name,Age,Course,Year).
- ii) Alter table.
- iii) Drop table.
- iv) Truncate table.

Write necessary query statements.

2. Execute DML statements (Select,Insert,Update,Delete)

- i) Create table Employee (Emp_Id,Emp_Name,Dept_Id,Salary) And Also create another table Department (Dept_Id,Dept_Name,Dept_Head)
- ii) Insert minimum of 4 rows.
- iii) Set Primary Key and Foreign Key constraints.
- iv) Display the records.
- v) Update a record.
- vi) Delete a record.

Write necessary query statements.

3. Create a table and execute DCL statements (grant,revoke).

4. Create a table and execute TCL statements (commit,rollback,savepoint).

5. Accessing database (SELECT, Filtering using WHERE, HAVING, GROUP BY, ORDER BY Clauses)

Consider the database (Employee) that contains the record Employee ID, Name, Department, Education, and their salary in Lacs.

Employee ID	Name	Gender	Department	Education	Month of Joining	Salary
1001	Ajay	M	Engineering	Doctoral	January	25
1002	Babloo	M	Engineering	UG	February	23
1003	Chhavi	F	HR	PG	March	15
1004	Dheeraj	M	HR	UG	January	12
1005	Evina	F	Marketing	UG	March	16
1006	Fredy	M	Sales	UG	December	10
1007	Garima	F	Sales	PG	March	10
1008	Hans	M	Admin	PG	November	8
1009	Ivanka	F	Admin	Intermediate	April	7
1010	Jai	M	Peon	High School	December	4

- 1) (i): Calculate the sum of salaries of each department.

(ii): Find the departments in which the SUM of the salaries is greater than or equal to 20 lacs

(iii) Display distinct department of Employee.

(iv) Display total number of Salary of Employees.

(v) Display details of Employees from Employee tables in which Department of the employee is Engineering and Education is Doctorial.

(vi) Display details of Employees from Employee tables in which Department of the employee is Engineering or Education is Doctorial.

(vii) Rename the columns "Name" and "Education" to "First_Name" and "Qualification", respectively.

(viii)List records of Employees whose names start with “A”.

2) Find the department in which SUM salary is greater than or equal to 20 lacs, but the education of employees is not UG.

3) Find the departments in which the SUM of the salaries is greater than or equal to 15 lacs and arrange the Salary in descending order.

6. (i) From the following table, create a view for those salespeople who belong to the city of New York.

Table: salesman

salesman_id	name	city	commission
5001	James Hoog	New York	0.15
5002	Nail Knite	Paris	0.13
5005	Pit Alex	London	0.11
5006	Mc Lyon	Paris	0.14
5007	Paul Adam	Rome	0.13
5003	Lauson Hen	San Jose	0.12

(ii) From the following table, create a view that counts the number of customers in each grade.

Table: customer

customer_id	cust_name	city	grade	salesman_id
3002	Nick Rimando	New York	100	5001
3007	Brad Davis	New York	200	5001
3005	Graham Zusi	California	200	5002
3008	Julian Green	London	300	5002
3004	Fabian Johnson	Paris	300	5006
3009	Geoff Cameron	Berlin	100	5003
3003	Jozy Altidor	Moscow	200	5007
3001	Brad Guzan	London	100	5005

(iii) From the following table, create a view to count the number of unique customers, compute the average and the total purchase amount of customer orders by each date.

Table: orders

ord_no	purch_amt	ord_date	customer_id	salesman_id
70001	150.5	2012-10-05	3005	5002
70009	270.65	2012-09-10	3001	5005
70002	65.26	2012-10-05	3002	5001
70004	110.5	2012-08-17	3009	5003
70007	948.5	2012-09-10	3005	5002
70005	2400.6	2012-07-27	3007	5001
70008	5760	2012-09-10	3002	5001
70010	1983.43	2012-10-10	3004	5006
70003	2480.4	2012-10-10	3009	5003
70012	250.45	2012-06-27	3008	5002
70011	75.29	2012-08-17	3003	5007
70013	3045.6	2012-04-25	3002	5001

(iv) From the order table, create a view to find the salespersons who issued orders on either August 17th, 2012 or October 10th, 2012. Return salesperson ID, order number and customer ID.

(v) From the salesman and order tables, create a view to find the salesperson who handles a customer who makes the highest order of the day. Return order date, salesperson ID, name.

7. Subquery

Table: employees

```
mysql> SELECT * FROM employees;
```

emp_id	emp_name	emp_age	city	income
101	Peter	32	Newyork	200000
102	Mark	32	California	300000
103	Donald	40	Arizona	1000000
104	Obama	35	Florida	5000000
105	Linklon	32	Georgia	250000
106	Kane	45	Alaska	450000
107	Adam	35	California	5000000
108	Macculam	40	Florida	350000
109	Brayan	32	Alaska	400000
110	Stephen	40	Arizona	600000
111	Alexander	45	California	70000

- Find employee detail whose id matches in a subquery:
- Find employee detail whose income is more than 350000 with the help of subquery.
- Find employee details with maximum income using a subquery.

8. Sailors(sid: integer, sname: string, rating: integer, age: real);
 Boats(bid: integer, bname: string, color: string);
 Reserves(sid: integer, bid: integer, day: date).

sailors				reserves			boats		
sid	sname	rating	age	sid	bid	day	bid	bname	color
22	Dustin	7	45.0	22	101	10/10/98	101	Interlake	blue
29	Brutus	1	33.0	22	102	10/10/98	102	Interlake	red
31	Lubber	8	55.5	22	103	10/8/98	103	Clipper	green
32	Andy	8	25.5	22	104	10/7/98	104	Marine	red
58	Rusty	10	35.0	31	102	11/10/98			
64	Horatio	7	35.0	31	103	11/6/98			
71	Zorba	10	16.0	31	104	11/12/98			
74	Horatio	9	35.0	64	101	9/5/98			
85	Art	3	25.5	64	102	9/8/98			
95	Bob	3	63.5	74	103	9/8/98			

- Count the number of distinct boat colors:
 - Find all information of sailors who have reserved boat number 101.
 - Find names of sailors who have reserved at least one boat.
 - Find names of sailors who have reserved a red boat and list in the order of their age.
 - Display boat names and the names of sailors who have sailed on them:
 - Find the ids and names of sailors who have reserved two different boats on the same day.
 - Find the ids of sailors who have reserved a red boat or a green boat.
 - Find the names of sailors who have reserved all boats
9. Write a PL/SQL procedure to display all fields from a table.
10. Write a PL/SQL procedure to add two numbers
11. Write a PL/SQL procedure to check whether a number is odd or even.
12. Write a PL/SQL procedure to find Factorial of a number
13. Write a PL/SQL procedure to find maximum of three values.
14. Write a PL/SQL procedure to find the sum of digits
15. SQL Trigger
 Execution of trigger

The database consists of three tables: "students", "courses", and "enrollment_log". The "students" table stores information about students, including their student ID, name, and email address. The "courses" table contains details about courses offered by the university, such as course ID, title, and instructor. The "enrollment_log" table records all enrollment transactions, tracking the enrollment of students into courses. Create a trigger named "after_enrollment" that automatically inserts a log entry into the "enrollment_log" table whenever a student enrolls in a course. The log entry should include details about the enrollment action (e.g., "Student enrolled in course"), along with the timestamp of the enrollment.

16. MySQL Trigger : BEFORE UPDATE

17. MySQL Trigger : AFTER UPDATE

18. Execution of cursor

Write a program in PL/SQL to find average salary using cursor

19. Write a program in PL/SQL to list the name of students using cursor

20. MongoDB

- i) Designing Databases using NoSQL
- ii) Create database test and Make Collection With name "student" (Rollno,Name,Branch).
- iii) Insert Records Into student Collection(10 records).
- iv) Display Data in Proper Format.

21. Designing NoSQL Database - Employee Management

- i) Create a NoSQL database named "Employee".
- ii) Create a collection named "EMPL" with fields: "Empno", "Name", "Salary", and "Role".
- iii) Insert 10 records into the "EMPL" collection.
- iv) Display the data from the "EMPL" collection in a proper format.
- v) Retrieve employees from the "EMPL" collection based on their roles.
- vi) Update the salary of an employee in the "EMPL" collection.

22. Performing CRUD Operations - Product Catalog

- i) Create a NoSQL database named "ProductCatalog".
- ii) Create a collection named "Products" with fields: "ProductID", "ProductName", "Price", and "Quantity".
- iii) Insert several records into the "Products" collection.
- iv) Display the data from the "Products" collection.
- v) Update the details of a specific product. (For example, increase the quantity of laptops by 10)
- vi) Delete a product from the collection. (For example, remove the smartphone from the catalog.)

23. Usage of aggregate functions

```
{
  "_id" : ObjectId("...")
  "name" : "Tony",
  "age" : 17,
  "id" : 1,
  "sec" : "A",
  "subject" : ["physics", "maths"]
}
{
  "_id" : ObjectId("...")
  "name" : "steve",
  "age" : 37,
  "id" : 2,
  "sec" : "A"
}
{
  "_id" : ObjectId("...")
  "name" : "natasha",
  "age" : 17,
  "id" : 3,
  "sec" : "B",
  "subject" : ["physics", "english"]
}
{
  "_id" : ObjectId("...")
  "name" : "bruse",
  "age" : 21,
  "id" : 4,
  "sec" : "B",
  "subject" : ["physics", "maths", "Biology", "chemistry"]
}
{
  "_id" : ObjectId("...")
  "name" : "nick",
  "age" : 40,
  "id" : 5,
  "sec" : "B",
  "subject" : ["english"]
}
{
  "_id" : ObjectId("...")
  "name" : "groot",
  "age" : 4,
  "id" : 6,
  "sec" : "A",
```

```

    "subject" : ["english"]
  }
  {
    "_id" : ObjectId("...")
    "name" : "thanos",
    "age" : 4,
    "id" : 7,
    "sec" : "A",
    "subject" : ["maths", "physics", "chemistry"]
  }

```

- i) Displaying the total number of students in one section only
- ii) Displaying the total number of students in both the sections and maximum age from both section
- iii) Displaying details of students whose age is greater than 30 using match stage
- iv) Sorting the students on the basis of age
- v) Counting the total numbers of documents
- vi) Displaying distinct names and ages (non-repeating)

24. Create users and roles in the employee database

Program No.	Date
1,2	6/2/2024
3,4	9/2/2024
5,	27/2/2024
6,7	1/3/2024
8	12/3/2024
9,10	15/3/2024
11,12	19/3/2024
13,14	22/3/2024
15,16	26/3/2024
17,18	5/4/2024
19,20	9/4/2024
21,22	12/4/2024
23,24	16/4/2024